Broadband Expansion and Small-Firm Productivity: Evidence from a Staggered Municipal Rollout

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Demonstration manuscript; fictional setting and simulated data.

Key question

Can municipal broadband availability raise small-firm productivity?

Why this matters:

- Small firms adopt new technologies later and less completely
- Distance can limit access to specialized providers and inputs

Empirical test:

- Is labor productivity higher after municipal service arrives?
- Is the difference larger for communication-intensive firms?

Policy relevance: whether universal infrastructure changes small-firm production.

This paper

We link 9,400 firms in 2008–2019 to broadband rollout across 284 municipalities.

1. **Each 2011–2015 rollout cohort is more productive after backbone arrival.**
   - Pooled two-way estimate: 0.034 log points, the low end of the cohort range
2. **Communication-intensive firms show a larger difference.**
   - One SD adds 0.0064 log points, about one-fifth of the pooled difference
   - The model links this pattern to the value of remote coordination

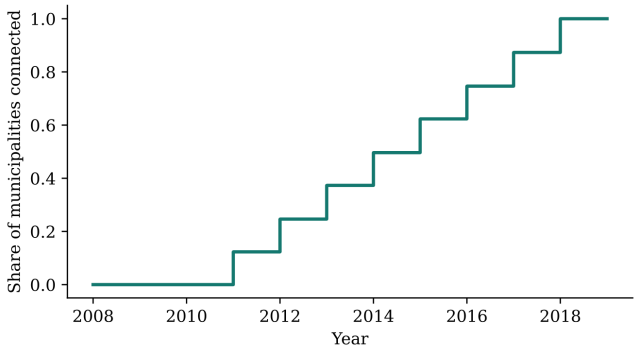
Implication: the productivity differences are largest where communication frictions bind.

Roadmap

1. Setting and data
2. Design and main evidence
3. Model and interpretation

Broadband reached every municipality in eight waves

All 284 municipalities connected between 2011–2018



- **Measurement:** exposure begins with municipal service; firm adoption is unobserved.

Data

Business Register, audited accounts, and employer–employee records, 2008–2019

- **Sample:** 9,400 firms and 112,800 firm-years, 2008–2019
  - 3–49 employees in 2008, positive sales, balanced survivor panel
- **Outcome:** real value added per worker
  - audited accounts deflated by industry prices, divided by FTE employment
- **Broadband:** first commercial-service year in the headquarters municipality
- **Communication intensity:** 2008 wage-bill share in communication-intensive jobs

▸ Sample details

The design follows firms as local backbone service arrives

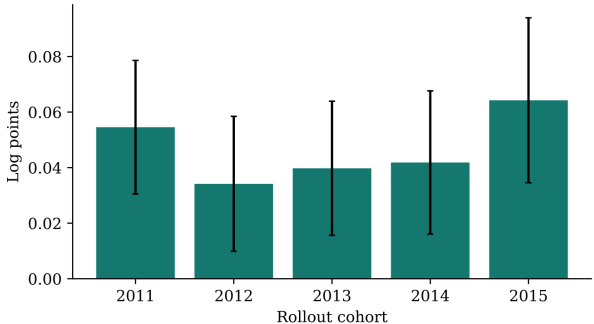
Within-firm comparison across different municipal arrival dates

$$y_{it} = \alpha_i + \lambda_t + \beta B_{m(i)t} + X'_{it}\delta + \varepsilon_{it}$$

- **Comparison:** each wave versus municipalities connected 3+ years later
- **Pooled summary:** one coefficient across all timing comparisons
- **Counterfactual:** municipalities served at different dates would follow similar paths
- **Diagnostic:** pre-arrival coefficients check for diverging trends before arrival

Each 2011–2015 cohort is more productive after arrival

Paper Figure 3; each cohort versus municipalities connected at least three years later

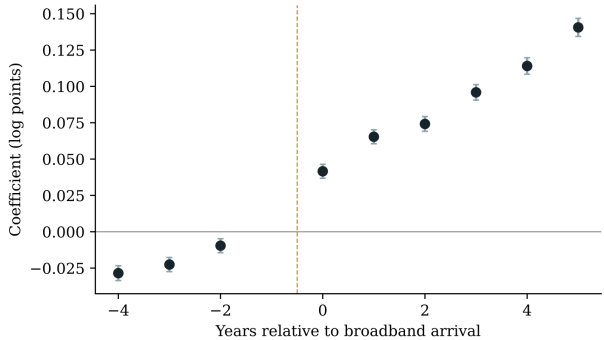


- **Estimator:** no already-connected municipality serves as a control here.

▸ Pooled estimate ▸ Source notes ▸ Inference

Productivity rises before and after broadband arrives

Paper Figure 2; two-way event study; $t = -1$ omitted; firm-clustered 95% intervals



- **Limitation:** The visible pretrend limits the estimates to a descriptive reading.

▸ Assignment ▸ Placebo

The model predicts larger gains for communication-intensive firms

Broadband can make specialized remote providers worthwhile

Switching case:

$$\kappa_1 p \leq \phi w < \kappa_0 p, \qquad G(\theta_i) = \eta \ln[1 + (\phi - 1)\theta_i]$$

$$G'(\theta_i) = \frac{\eta(\phi - 1)}{1 + (\phi - 1)\theta_i} > 0$$

- θ_i is firm i's pre-program communication intensity
- κ is the cost of coordinating with remote providers
- $\phi > 1$ is the quality advantage of specialized remote providers

Prediction: communication-intensive firms can shift more tasks to remote providers.

▸ Derivation

The broadband association is larger for communication-intensive firms

Communication intensity is fixed at its 2008 value

	With interaction	Average only
Broadband	0.020***	0.034***
	(0.003)	(0.002)
Broadband × communication intensity	**0.032*** **	
	(0.005)	
Firm and year effects	Yes	Yes
Observations	112,800	112,800
One SD: $0.032 \times 0.20 =$ **0.0064** log points		

Firm-clustered SEs; *** $p < 0.01$ as in the source; communication-intensity SD = 0.20 (Table 1).

- The larger gap for communication-intensive firms matches the model.

▸ Interpretation

Conclusion

- **Each 2011–2015 cohort is more productive after broadband arrives.**
- The difference is larger for communication-intensive firms.

Implication: the productivity differences are largest where communication frictions bind.

Appendix

The balanced panel describes surviving small firms

Paper Section 4.2 and Table 1, p. 5

- **Population:** registered firms linked to audited accounts and worker records
- **Initial screen:** 3–49 employees in 2008; positive sales; at least four consecutive years
- **Final restriction:** a balanced 2008–2019 panel
- **Delivered sample:** 9,400 firms and 112,800 firm-years
- **Scope:** the comparison applies to selected survivors; entry and exit are excluded

Source issues: balance rule unexplained; reported employment mean 23.4 exceeds maximum 19.

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Projected growth helped determine rollout order

Published RDDF priority rule, paper Section 4.1

- **Coverage commitment:** the 2010 Act promised universal connection by 2018
- **Priority rule:** earlier service for low-productivity, high-growth municipalities
  - below-median productivity levels
  - above-median projected growth in internal assessments
- **Engineering constraints:** distance and terrain also affected construction
- **Unresolved:** the paper does not isolate this variation from the growth score

Identification requires: arrival timing unrelated to counterfactual productivity trends.

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The pooled two-way estimate sits at the low end

Paper Table 2, p. 7; log labor productivity, firm and year fixed effects

	Baseline	Firm controls	Region trends	Municipal SEs
Broadband	**0.034*** **	0.033***	0.030***	0.034***
	(0.002)	(0.002)	(0.002)	(0.003)
Firm and year effects	Yes	Yes	Yes	Yes
Observations	112,800	112,800	112,800	112,800

SEs are firm-clustered except the final column (municipality-clustered); *** $p < 0.01$, as in the source.

- **Estimator:** a weighted mix of all timing comparisons, it can understate rising effects.

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Rounded placebo entries cannot confirm the prose

Pre-treatment sample; rollout years permuted across municipalities

	Pre-period placebo
Permuted broadband	−0.004
	(0.002)

- The coefficient-to-SE ratio is about −2; rounding allows −1.4 to −3
- The prose calls the estimate statistically indistinguishable from zero
- The source provides no exact *p*-value or confidence interval

- **Unresolved:** the rounded entries fit both a zero and a nonzero reading.

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Two numerical inconsistencies require correction

Narrative prose versus exact table cells

Quantity	Narrative source	Exact table source
Headline magnitude	Abstract: 34%; text: 3.4%	Table 2: 0.034 log points
Region-trend estimate	Section 6.1: 0.024	Table 2: 0.030

- The talk reports the coefficient's table unit: log points
- The pooled-estimate backup uses exact Table 2 cells in every specification
- A 0.034 log-point difference is approximately 3.4%, not 34%

- **Correction:** the exact cell is 0.034 log points; both narrative values need revision.

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Table 5 is stable but not self-consistent

Exact reported cells, paper p. 9

	Region trends	No controls	Drop 2011–12
Broadband	0.030***	0.034***	0.037***
	(0.002)	(0.002)	(0.002)

- **Stable pattern:** reported coefficients range from 0.030 to 0.037
- **Header–note conflict:** "No controls" versus "full control set"
- **Prose–table conflict:** 0.024 versus 0.030 for region trends

- **Reading:** the coefficient range is narrow, but the source labels need correction.

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Municipal assignment calls for municipal-level inference

Treatment timing varies across 284 municipalities

	Firm-clustered	Municipality-clustered
Broadband	0.034	0.034
	(0.002)	(0.003)

- The baseline clusters by firm, below the municipal assignment level
- Table 2 column (4) reports the assignment-aligned alternative
- The point estimate is unchanged and the reported SE rises to 0.003

- **Inference:** clustering changes uncertainty, not the concern about selected timing.

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Larger gains require tasks to switch providers

Paper Proposition 1 and proof, pp. 2–3

$$\underbrace{\kappa_1 p}_{\text{remote cost after}} \leq \underbrace{\phi w}_{\text{quality-adjusted in-house cost}} < \underbrace{\kappa_0 p}_{\text{remote cost before}}$$

$$G(\theta_i) = \eta \ln[1 + (\phi - 1)\theta_i], \qquad G'(\theta_i) = \frac{\eta(\phi - 1)}{1 + (\phi - 1)\theta_i} > 0$$

- Before arrival, communication-intensive tasks remain in-house
- After arrival, the same tasks switch to specialized remote providers
- The derivative is positive because $\eta > 0$ and $\phi > 1$
- **Scope:** the heterogeneity prediction is conditional on this switch

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The interaction is consistent with, not unique to, outsourcing

What Table 4 observes and what it leaves unmeasured

- **Observed:** a positive interaction with 2008 communication intensity
  - one SD adds 0.0064 log points
- **Predicted:** the task model gives the same positive sign
  - the switching case makes $G'(\theta_i) > 0$
- **Unobserved:** remote procurement, task reorganization, and channel-specific inputs
- **Alternative:** other firm traits can generate the same pattern

Discriminating evidence: direct measures of sourcing before and after arrival.

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The six-year payback is not traceable from the reported inputs

Paper Section 7, pp. 8–9

- **Claimed flow:** 410 million euros per year against 2.3 billion euros in construction
- **Estimate:** a productivity association among sample survivors
- **Missing inputs:** firm value added, program coverage, and maintenance costs
- **Missing assumptions:** causality, persistence, discounting, and incidence
- **Unit problem:** gross output is not a fiscal return

Conclusion for this talk: the payback and cost-effectiveness claims are excluded.

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